

MILDENHALL, United Kingdom

WMO: 035770

Lat: 52.367N

Lon: 0.483E

Elev: 33

StdP: 14.68

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 35046

Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |             |             |             |             |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|-------------|-------------|-------------------------------|-------------|-------------|-------------|-------------|-------------|-----------------------|-------------|-------------|-------------|-----------------------|------------|--------------|
|               |             |             | 99.6%                         |             |             | 99%         |             |             | 0.4%                  |             | 1%          |             |                       |            |              |
|               | 99.6%       | 99%         | DP                            | HR          | MCDB        | DP          | HR          | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>25.0 | (c)<br>28.0 | (d)<br>21.4                   | (e)<br>16.1 | (f)<br>29.7 | (g)<br>24.5 | (h)<br>18.6 | (i)<br>31.2 | (j)<br>32.0           | (k)<br>49.5 | (l)<br>29.0 | (m)<br>49.0 | (n)<br>5.2            | (o)<br>240 | (p)<br>0.605 |

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD<br>to 0.4% DB |            |
|---------------|---------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|-------------------------|------------|
|               |                           | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                         |            |
|               |                           | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                    | PCWD       |
| (a)<br>7      | (b)<br>17.9               | (c)<br>82.7     | (d)<br>66.6 | (e)<br>78.9 | (f)<br>65.1 | (g)<br>75.4 | (h)<br>63.7 | (i)<br>69.0         | (j)<br>77.6 | (k)<br>67.2 | (l)<br>75.3 | (m)<br>65.6 | (n)<br>72.6 | (o)<br>10.2             | (p)<br>140 |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 66.4                            | 97.3 | 70.3 | 64.6 | 91.3 | 69.0 | 62.9 | 86.0 | 67.5 | 33.1          | 78.4 | 31.6 | 75.0 | 30.4 | 73.0 | 77.2              |

Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |      |                    | n-Year Return Period Values of Extreme Temperature |           |      |            |      |            |      |            |     |      |
|-------------------|------|------|----------------------------|------|------|--------------------|----------------------------------------------------|-----------|------|------------|------|------------|------|------------|-----|------|
| 1%                |      | 2.5% | 5%                         | Mean |      | Standard Deviation |                                                    | n=5 years |      | n=10 years |      | n=20 years |      | n=50 years |     |      |
| Min               |      | Max  |                            | Min  | Max  |                    | Min                                                | Max       | Min  | Max        | Min  | Max        | Min  | Max        |     |      |
| (a)               | (b)  | (c)  |                            | (e)  | (f)  | (g)                | (h)                                                | (i)       | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p) |      |
| 25.7              | 22.9 | 20.4 |                            | DB   | 20.0 | 89.6               | 4.0                                                | 3.8       | 17.2 | 92.4       | 14.8 | 94.6       | 12.6 | 96.7       | 9.7 | 99.5 |
|                   |      |      |                            | WB   | 19.7 | 72.5               | 4.1                                                | 1.9       | 16.8 | 73.8       | 14.4 | 74.9       | 12.2 | 75.9       | 9.2 | 77.3 |

Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 51.6        | 40.4  | 41.4  | 44.6  | 49.2  | 54.7  | 60.0  | 64.6  | 64.0  | 59.0  | 53.0  | 45.8  | 41.4  | (6)  |
|                                                                        | DBStd       | 10.25       | 6.07  | 6.12  | 5.60  | 5.89  | 5.68  | 4.98  | 5.01  | 4.82  | 4.81  | 5.62  | 6.02  | 6.81  | (7)  |
|                                                                        | HDD50       | 1289        | 299   | 246   | 180   | 84    | 20    | 0     | 0     | 0     | 1     | 33    | 150   | 276   | (8)  |
|                                                                        | HDD65       | 5035        | 761   | 660   | 631   | 473   | 323   | 165   | 71    | 78    | 190   | 374   | 576   | 733   | (9)  |
|                                                                        | CDD50       | 1863        | 3     | 5     | 14    | 62    | 165   | 300   | 452   | 433   | 270   | 125   | 25    | 9     | (10) |
|                                                                        | CDD65       | 130         | 0     | 0     | 0     | 0     | 2     | 14    | 58    | 46    | 9     | 1     | 0     | 0     | (11) |
|                                                                        | CDH74       | 1092        | 0     | 0     | 0     | 5     | 31    | 146   | 496   | 341   | 66    | 7     | 0     | 0     | (12) |
|                                                                        | CDH80       | 257         | 0     | 0     | 0     | 0     | 1     | 27    | 124   | 91    | 13    | 1     | 0     | 0     | (13) |
| (14) Wind                                                              | WSAvg       | 9.9         | 11.5  | 11.3  | 11.0  | 10.0  | 9.8   | 9.1   | 8.7   | 8.7   | 8.8   | 9.7   | 9.9   | 10.7  | (14) |
| (15) Precipitation                                                     | PrecAvg     | 23.1        | 2.0   | 1.5   | 1.3   | 1.5   | 1.8   | 2.0   | 2.1   | 2.3   | 1.9   | 2.3   | 2.3   | 1.9   | (15) |
|                                                                        | PrecMax     | 30.2        | 3.8   | 2.9   | 3.1   | 4.5   | 4.8   | 5.8   | 4.1   | 5.0   | 4.3   | 4.4   | 4.3   | 3.7   | (16) |
|                                                                        | PrecMin     | 15.3        | 0.5   | 0.2   | 0.1   | 0.0   | 0.4   | 0.6   | 0.8   | 0.2   | 0.3   | 0.5   | 1.0   | 0.7   | (17) |
|                                                                        | PrecStd     | 4.0         | 0.8   | 0.8   | 0.7   | 1.1   | 1.0   | 1.3   | 0.9   | 1.1   | 1.1   | 1.0   | 0.9   | 0.8   | (18) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 56.8  | 59.2  | 63.6  | 73.4  | 77.3  | 83.6  | 88.2  | 87.5  | 81.0  | 71.3  | 62.2  | 57.7  | (19) |
|                                                                        |             | MCWB        | 53.6  | 54.4  | 53.4  | 58.9  | 62.9  | 66.9  | 67.7  | 67.9  | 65.3  | 62.8  | 59.3  | 54.7  | (20) |
|                                                                        | 2%          | DB          | 54.0  | 55.4  | 59.7  | 66.6  | 73.5  | 78.5  | 83.1  | 82.3  | 74.9  | 66.3  | 59.1  | 55.7  | (21) |
|                                                                        |             | MCWB        | 51.0  | 51.6  | 52.7  | 56.0  | 61.5  | 64.6  | 66.7  | 66.8  | 63.5  | 60.8  | 56.0  | 53.3  | (22) |
|                                                                        | 5%          | DB          | 51.9  | 53.2  | 57.0  | 62.9  | 69.8  | 73.8  | 79.1  | 77.2  | 71.2  | 63.8  | 57.1  | 53.8  | (23) |
|                                                                        |             | MCWB        | 49.2  | 49.6  | 51.3  | 54.2  | 59.7  | 62.7  | 65.3  | 64.8  | 62.0  | 59.2  | 54.4  | 51.6  | (24) |
|                                                                        | 10%         | DB          | 49.9  | 50.3  | 53.9  | 59.4  | 66.0  | 70.2  | 75.5  | 73.7  | 67.9  | 61.2  | 55.1  | 51.7  | (25) |
|                                                                        |             | MCWB        | 47.7  | 47.8  | 49.4  | 52.5  | 57.6  | 61.0  | 64.0  | 62.9  | 60.3  | 57.5  | 52.6  | 49.6  | (26) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 54.3  | 54.9  | 56.9  | 60.8  | 67.5  | 70.2  | 71.1  | 70.9  | 68.5  | 65.1  | 59.6  | 56.0  | (27) |
|                                                                        |             | MCDB        | 55.8  | 57.5  | 60.1  | 68.7  | 72.6  | 79.0  | 79.4  | 80.2  | 76.1  | 68.8  | 60.9  | 57.0  | (28) |
|                                                                        | 2%          | WB          | 52.1  | 52.3  | 54.4  | 58.0  | 63.4  | 67.3  | 68.9  | 68.4  | 65.6  | 62.3  | 56.9  | 54.2  | (29) |
|                                                                        |             | MCDB        | 54.1  | 54.8  | 57.9  | 64.1  | 69.6  | 73.5  | 78.3  | 77.0  | 71.6  | 65.3  | 58.5  | 55.7  | (30) |
|                                                                        | 5%          | WB          | 49.9  | 50.3  | 52.4  | 55.8  | 60.8  | 64.5  | 67.1  | 66.6  | 63.8  | 60.2  | 54.9  | 52.2  | (31) |
|                                                                        |             | MCDB        | 51.6  | 52.2  | 55.8  | 61.3  | 68.1  | 70.9  | 76.1  | 74.5  | 68.8  | 62.6  | 56.8  | 53.8  | (32) |
|                                                                        | 10%         | WB          | 47.8  | 48.4  | 50.4  | 53.5  | 58.5  | 62.6  | 65.4  | 65.0  | 62.0  | 58.6  | 52.8  | 49.9  | (33) |
|                                                                        |             | MCDB        | 49.4  | 50.3  | 53.6  | 58.9  | 64.5  | 68.6  | 73.2  | 71.5  | 66.1  | 60.9  | 54.7  | 51.3  | (34) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 10.0  | 11.5  | 14.3  | 16.6  | 16.9  | 17.1  | 17.9  | 17.5  | 16.4  | 13.8  | 11.2  | 9.5   | (35) |
|                                                                        |             | MCDBR       | 11.6  | 13.1  | 18.3  | 21.1  | 22.9  | 23.1  | 24.4  | 22.7  | 20.3  | 15.0  | 11.2  | 10.0  | (36) |
|                                                                        | 5% WB       | MCWBR       | 10.7  | 10.2  | 12.2  | 12.3  | 12.8  | 11.8  | 10.2  | 9.8   | 10.9  | 10.2  | 9.7   | 9.4   | (37) |
|                                                                        |             | MCWBR       | 11.2  | 12.2  | 15.7  | 19.1  | 20.3  | 19.9  | 21.0  | 19.3  | 17.4  | 13.8  | 11.3  | 10.1  | (38) |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.327 | 0.344 | 0.378 | 0.403 | 0.407 | 0.411 | 0.410 | 0.399 | 0.377 | 0.358 | 0.331 | 0.316 | (40) |
|                                                                        |             | taud        | 2.357 | 2.372 | 2.267 | 2.215 | 2.247 | 2.258 | 2.273 | 2.325 | 2.384 | 2.420 | 2.404 | 2.345 | (41) |
|                                                                        | Ebn at Noon |             | 210   | 243   | 255   | 263   | 268   | 267   | 265   | 262   | 254   | 234   | 207   | 191   | (42) |
|                                                                        |             | Edh at Noon | 19    | 25    | 34    | 40    | 41    | 41    | 40    | 36    | 30    | 24    | 18    | 17    | (43) |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 233   | 448   | 800   | 1260  | 1526  | 1627  | 1554  | 1307  | 981   | 571   | 289   | 185   | (44) |
|                                                                        | RadStd      |             | 22    | 66    | 90    | 131   | 115   | 127   | 160   | 94    | 81    | 52    | 31    | 23    | (45) |

Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | +1.12 | N/A         | N/A   | N/A   | N/A   |
| Regional (80 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page